

strong, stable, covalent bonds with many other types of atoms with higher or lower values. When carbon atoms bond with different types of atoms, they form molecules with properties that vary according to the atoms they've bonded with.

2. Each carbon atom can form these covalent bonds with four other atoms at a time. That's more bonds than most other atoms can form. Each nitrogen atom (for example) can form only three covalent bonds, each oxygen atom can form only two covalent bonds, and so on. This four-bond capability allows carbon atoms to bond to other carbon atoms to make chains of atoms – and to bond with other kinds of atoms at various points along such chains.
3. There's no other element in the periodic table that bonds as strongly to itself and in as many ways as the carbon atom. Carbon atoms can bond together in short chains, in which case they may have the properties of a gas. They may bond together as long chains, which might give you a solid material, like plastic. Or, they can bond together in 2- or 3-dimensional lattices, which can make for some very hard materials, such as diamonds.

Hence it's only natural that carbon atoms will find uses in nanomaterials.

Classification: One dimension

Thin films, layers and surfaces

One-dimensional nanomaterials, such as thin films and engineered surfaces, have been developed and used for decades in fields such as electronic device manufacturing, chemistry and engineering. If you went to a silicon integrated-circuit industry, for example, you'll witness many devices relying on thin films for their operation, and control of film thicknesses approaching the atomic level is routine. Monolayers (layers that are one atom or molecule deep) are also routinely made and used in chemistry. The formation and properties of these layers are reasonably well understood from the atomic level upwards, even in quite complex layers (such as lubricants). Advances are being made in the control of the composition and smoothness of surfaces and the growth of films.

Engineered surfaces with tailored properties such as large surface area or specific reactivity are used routinely in a range of applications such as in fuel cells and catalysts. The large surface area provided by nanoparticles, together with their ability to self-assemble on a support surface, could be of use in all of these applications.

Although they represent incremental developments, surfaces with enhanced properties should find applications throughout the chemicals and energy sectors. The benefits could surpass the obvious economic and resource savings